The Mercedes-star partition of a disk

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October 12, 2026

Abstract

Among triples of pairwise disjoint nonempty connected open subsets of the unit disk, the three sectors of opening $2\pi/3$ minimize the largest first Dirichlet eigenvalue. The minimum is $j_{3/2,1}^2$, where $j_{3/2,1}$ is the first positive zero of the Bessel function $J_{3/2}$. The proof transfers any triple with smaller energy to a triple on the unit sphere with energy below its known sharp minimum 15/4. A common radial change of variables matches the angular terms after division by positive reference profiles. A Wronskian comparison shows that the remaining radial term has the required sign. The argument imposes no condition on the cells' boundary regularity, topology, symmetry, or coverage of the disk.

1 The minimax problem

Spectral partition problems balance the first Dirichlet eigenvalues of several disjoint cells. For three cells in a disk, the natural candidate is the Mercedes star: three equal sectors meeting at the center. Let $B = \{x \in \mathbb{R}^2 : |x| < 1\}$. For a nonempty open subset D of B, write

$$\lambda_1(D) = \inf_{0 \neq u \in C_c^\infty(D;\mathbb{R})} \frac{\int_D |\nabla u|^2}{\int_D u^2}. \tag{1}$$

This also equals the infimum over $H_0^1(D)$, defined as the closure of $C_c^\infty(D)$ in $H^1(D)$: convergence in that norm passes the energy inequality to the closure, and the reverse comparison follows from inclusion. This convention is valid for arbitrary open sets.

Berkolaiko, Canzani, Cox and Marzuola prove minimality of the Mercedes star among partitions whose boundary contains a curve from the origin to the boundary of the disk [1, Theorem 1.8]. The unrestricted class also allows interfaces outside that comparison. On the unit sphere, in contrast, the three equal lunes are known to minimize the same objective [2, Theorem 1.1 and (1.9)]. The arbitrary-open-cell formulation is stated by Soave and Terracini [3, Definition 1 and Theorem 1.10]. We use that spherical result to obtain the unrestricted disk comparison.

Theorem 1. *For every triple (D_1, D_2, D_3) of pairwise disjoint nonempty connected open subsets of B,*

$$\max_{1 \le i \le 3} \lambda_1(D_i) \ge j_{3/2,1}^2.$$

The three sectors separated by radii at angles $2\pi/3$ attain equality. Thus the infimum over all such triples is $j_{3/2,1}^2$. Here $j_{3/2,1}$ is the first positive zero of the Bessel function $J_{3/2}$.

The comparison acts on arbitrary compactly supported test functions. We divide out the positive radial profile of an equal sector, and choose a radial map so that its angular energy agrees

with the corresponding spherical energy. The radial energy is then larger. Consequently, three disk cells with eigenvalues below the proposed value would yield three spherical cells below their sharp value. A logarithmic cutoff handles tests that meet the center. We prove the radial comparison in Section 2, the energy identity in Section 3, and the theorem in Section 4.

We record the external spectral result in the precise form used below. The sphere has radius one and its induced metric; its eigenvalues are defined by (1) with spherical gradient and area measure.

Proposition 2 (Spherical three-partition theorem)**.** *For every triple of pairwise disjoint nonempty connected open subsets* Ω_i *of* $\mathbb{S}^2$*, one has* $\max_i \lambda_1(\Omega_i) \geq 15/4$*.*

This is the $N = 3$ case of [3, Theorem 1.10], with the partition class of Definition 1 and the objective in equation (10) of that paper's arXiv v2. It requires neither a covering nor regular boundaries.

2 A radial comparison

Put $\nu = 3/2$ and $E = j_{3/2,1}^2$. It is useful to write the positive reference profile without Bessel notation. Let k be the first positive zero of $f(x) = \sin x - x\cos x$. Since $f' = x\sin x$, it is positive on $(0, \pi]$ and strictly decreases from π to -1 on $(\pi, 3\pi/2)$. Hence $k \in (\pi, 3\pi/2)$. The identities for the spherical Bessel function

$$j_1(z) = \sqrt{\frac{\pi}{2z}} J_{3/2}(z) = \frac{\sin z}{z^2} - \frac{\cos z}{z} \quad (z > 0)$$

[4, 10.47.3 and 10.49.3] identify $k = j_{3/2,1}$. Define

$$q(r) = r^{-3/2}\{\sin(kr) - kr\cos(kr)\}, \qquad 0 < r < 1. \tag{2}$$

It is positive and solves

$$-q'' - \frac{q'}{r} + \frac{\nu^2}{r^2} q = Eq. \tag{3}$$

Indeed, for $g(r) = f(kr)$, $g' = k^2 r\sin(kr)$ and $g'' = k^2\sin(kr) + k^3 r\cos(kr)$; the left side is $r^{-3/2}(2g'/r - g'') = k^2 q$. At the endpoints,

$$\begin{aligned} q(r) &= ar^{3/2} + O(r^{7/2}), \quad q'(r) = \tfrac{3}{2} ar^{1/2} + O(r^{5/2}), \\ a &= k^3/3 > 0, \qquad\quad q(1) = 0, \qquad q'(1) = k^2 \sin k < 0. \end{aligned} \tag{4}$$

The spherical profile will be $\sin^{3/2}\theta$. To match the angular energy after dividing by these profiles, set

$$\begin{aligned} T &= \int_0^1 q(r)^2 \frac{dr}{r}, \qquad C = \frac{2T}{\pi}, \\ \int_0^{\theta(r)} \sin^2 a \, da &= \frac{1}{C} \int_0^r q(t)^2 \frac{dt}{t}. \end{aligned} \tag{5}$$

Here $0 < T < \infty$ by (4). The cumulative functions increase strictly from 0 to $\pi/2$; therefore $\theta : (0, 1) \to (0, \pi)$ is a smooth increasing bijection, with endpoint limits 0 and π. In particular,

$$\theta' = \frac{q^2}{Cr\sin^2\theta}. \tag{6}$$

Lemma 3. *For every* $0 < r < 1$,

$$q(r)^2 > C \sin^3 \theta(r), \qquad h(r) := \frac{r\theta'(r)}{\sin \theta(r)} > 1. \tag{7}$$

Proof. Use logarithmic radius: $Q(t) = q(e^t)$ on $t < 0$, extended by zero on $t \geq 0$. For each real s, compare it with $Z_s(t) = \sqrt{C}\,\operatorname{sech}^{3/2}(t - s)$ on $\mathbb{R}$. The substitution $a = 2 \arctan e^{t-s}$ gives $a' = \sin a = \operatorname{sech}(t - s)$, and hence

$$\int_{\mathbb{R}} Q^2\,dt = T = \int_{\mathbb{R}} Z_s^2\,dt.$$

On $t < 0$, (3) and differentiation give

$$Q'' = (\nu^2 - Ee^{2t})Q, \qquad Z_s'' = (\nu^2 - \nu(\nu + 1)\operatorname{sech}^2(t - s))Z_s.$$

The Wronskian $W = Q'Z_s - QZ_s'$ therefore satisfies

$$W' = e^{2t} b_s(t) Q Z_s, \qquad b_s(t) = \frac{15e^{-2s}}{(1 + e^{2(t-s)})^2} - E. \tag{8}$$

The coefficient b_s decreases strictly, and $b_s(0) \leq 15/4 - E < 0$. Also $W(-\infty) = 0$, because both profiles and their derivatives are $O(e^{3t/2})$, while $W(0^-) = q'(1)Z_s(0) < 0$. If b_s is negative throughout, then $W < 0$. Otherwise W first increases from zero and then strictly decreases, crossing zero once before $t = 0$. Since $(Q/Z_s)' = W/Z_s^2$, the ratio is strictly decreasing, or increases and then decreases. Its zero extension adds no peak.

It follows that $I = \{Q^2 > Z_s^2\}$ is a single open interval. It is nonempty, since the two total masses agree and $Q^2 - Z_s^2 < 0$ on $[0, \infty)$. Its right endpoint b is less than zero. Write $H(t) = \int_{-\infty}^{t} (Q^2 - Z_s^2)\,da$. If $I = (-\infty, b)$, then $H(t) > 0$ at every finite t: before b it is an integral of a positive function, and after b it is minus the remaining negative tail integral. If $I = (a, b)$ with finite a, then $H < 0$ up to a, H is strictly increasing on (a, b), and

$$H(b) = -\int_b^{\infty} (Q^2 - Z_s^2)\,dt > 0.$$

Outside I the density difference is strictly negative except at its endpoints; strict monotonicity of the ratio excludes an equality interval. Thus any finite zero of H lies strictly inside I. We have proved

$$\int_{-\infty}^{t_0} Q^2\,dt = \int_{-\infty}^{t_0} Z_s^2\,dt \quad \Longrightarrow \quad Q(t_0)^2 > Z_s(t_0)^2 \qquad (t_0 < 0). \tag{9}$$

For a fixed r, choose $t_0 = \log r$ and $s = t_0 - \log \tan(\theta(r)/2)$. The substitution defining a shows that (5) is exactly the cumulative equality in (9). Since $\operatorname{sech}(t_0 - s) = \sin \theta(r)$, this gives the first inequality in (7); the second follows from (6). □

3 Comparison of quadratic forms

The radial map

$$\Phi(r, \phi) = (\sin \theta(r) \cos \phi, \sin \theta(r) \sin \phi, \cos \theta(r)), \qquad \Phi(0) = (0, 0, 1), \tag{10}$$

is a homeomorphism from B onto the sphere minus its south pole. Indeed, the radial functions are inverse continuous increasing bijections, and the angular circles collapse continuously at the origin

and north pole. Away from these two points, it is a smooth diffeomorphism. Its image is open in the full sphere, so it preserves openness, connectedness, nonemptiness and disjointness of cells.

Let u be a real compact smooth test on B whose support avoids the origin. Set

$$u = qv, \qquad V(\theta,\phi) = v(r(\theta),\phi), \qquad w(\theta,\phi) = \sin^{\nu}\theta\, V(\theta,\phi).$$

Extend w by zero near the two poles. It is smooth there because its support stays away from both poles, and it is nonzero whenever u is. Define the shifted forms

$$G_B(u) = \int_B |\nabla u|^2 - E\int_B u^2, \qquad G_S(w) = \int_{\mathbb{S}^2} |\nabla_S w|^2 - \tfrac{15}{4}\int_{\mathbb{S}^2} w^2.$$

Expanding $u_r = q'v + qv_r$ and integrating the mixed term gives

$$G_B(u) = \int_0^{2\pi}\int_0^1 \left[rq^2 v_r^2 + \frac{q^2}{r}(v_\phi^2 - \nu^2 v^2)\right] dr\, d\phi. \tag{11}$$

For clarity, the coefficient of v^2 after that integration is $rq'^2 - Erq^2 - (rqq')' = -q(rq')' - Erq^2 = -\nu^2 q^2/r$. All boundary terms vanish by compact support.

On the sphere the area measure is $\sin\theta\, d\theta\, d\phi$ and $|\nabla_S w|^2 = w_\theta^2 + w_\phi^2/\sin^2\theta$. The profile $p(\theta) = \sin^{\nu}\theta$ satisfies

$$-p'' - \cot\theta\, p' + \frac{\nu^2}{\sin^2\theta}p = \nu(\nu+1)p = \tfrac{15}{4}p.$$

This follows from $p'/p = \nu\cot\theta$ and $\csc^2\theta - \cot^2\theta = 1$. The corresponding mixed-term integration yields

$$G_S(w) = \int_0^{2\pi}\int_0^{\pi} \left[\sin^4\theta\, V_\theta^2 + \sin^2\theta\,(V_\phi^2 - \nu^2 V^2)\right] d\theta\, d\phi. \tag{12}$$

There are again no boundary terms, since the support avoids the poles. By (6) and (7),

$$\frac{q^2}{r\theta'} = C\sin^2\theta, \qquad rq^2\theta' = C\sin^4\theta\, h^2.$$

Changing variables in (11) and subtracting C times (12) proves the exact identity

$$G_B(u) - CG_S(w) = C\int_0^{2\pi}\int_0^{\pi} \sin^4\theta\,[h(r(\theta))^2 - 1]V_\theta^2\, d\theta\, d\phi \geq 0. \tag{13}$$

Thus a negative disk form gives a negative spherical form. In particular, no comparison of the two unweighted masses is needed.

4 Proof of Theorem 1

Suppose that an admissible triple had $\lambda_1(D_i) < E$ for all three indices. Choose nonzero $u_i \in C_c^\infty(D_i;\mathbb{R})$ with $G_B(u_i) < 0$. We first make their supports avoid the origin without losing these strict inequalities. Choose a smooth $\eta : \mathbb{R} \to [0,1]$, zero on $(-\infty, 0]$ and one on $[1,\infty)$, and put

$$\chi_\varepsilon(r) = \eta\left(\frac{\log r - 2\log\varepsilon}{\log(1/\varepsilon)}\right), \qquad 0 < \varepsilon < 1/2.$$

It vanishes for $r \leq \varepsilon^2$ and equals one for $r \geq \varepsilon$. Its constant extension near zero is smooth. With $M = \|\eta'\|_\infty$,

$$\int_B |\nabla\chi_\varepsilon|^2 \leq \frac{2\pi M^2}{\log(1/\varepsilon)} \longrightarrow 0.$$

For a fixed compact smooth u, the L^2 errors $(\chi_\varepsilon - 1)u$ and $(\chi_\varepsilon - 1)\nabla u$ are bounded by their supremum norms times $\sqrt{\pi}\varepsilon$, while $\|u\nabla\chi_\varepsilon\|_2 \le \|u\|_\infty\|\nabla\chi_\varepsilon\|_2 \to 0$. Hence $\chi_\varepsilon u \to u$ in $H^1(B)$. A common sufficiently small ε leaves all three products nonzero with negative G_B, still compactly supported in their original cells. Relabel them u_i.

The cells $\Omega_i = \Phi(D_i)$ are pairwise disjoint nonempty connected open spherical sets. The tests obtained in Section 3 are nonzero members of $C_c^\infty(\Omega_i)$, with $G_S(w_i) < 0$ by (13). Thus all three $\lambda_1(\Omega_i) < 15/4$, contradicting Proposition 2. This proves the universal lower bound.

For the upper bound, let $\ell = 2\pi/3$ and consider the sector $0 < r < 1$, $0 < \phi < \ell$. Its trial function $U = q(r)\sin(\nu\phi)$ has finite mass and gradient energy by (4). Multiplication of (3) by rq gives

$$\int_0^1 \left(rq'^2 + \nu^2 q^2/r\right)\, dr = E\int_0^1 rq^2\, dr,$$

since $rqq' \to 0$ at both endpoints. Both squared angular factors integrate to $\ell/2$, so the quotient of U is E.

To verify $U \in H_0^1$ explicitly, choose a smooth ρ bounded between zero and one, zero on $(-\infty, 1]$ and one on $[2, \infty)$. Multiply U by

$$\rho(r/\delta)\rho((1-r)/\delta)\rho(\phi/\delta)\rho((\ell-\phi)/\delta).$$

For small $\delta > 0$ this is a compact smooth sector test. The original mass and gradient integrals over the removed strips tend to zero by integrability. The extra derivative energies are $O(\delta^3)$ at the vertex, since $q = O(r^{3/2})$, and $O(\delta)$ at the circular edge, since $q = O(1-r)$. At either angular edge the sine vanishes linearly, and the extra energy is bounded by

$$K\left(\int_0^1 q^2\frac{dr}{r}\right)\delta^{-2}\int_0^{2\delta} a^2\, da = O(\delta).$$

Here K is independent of δ; the other cutoffs are bounded by one. The square of the sum of four extra gradient terms is at most four times the sum of their squares. The tests therefore converge to U in H^1. Their quotients tend to E, giving the sector eigenvalue at most E. The three rotated sectors are an admissible triple. Their maximum is at most E and, by the lower bound, at least E. This proves attainment and the stated exact infimum. □

References

[1] G. Berkolaiko, Y. Canzani, G. Cox and J. L. Marzuola, *Homology of spectral minimal partitions*, arXiv:2406.04225v1 (2024). https://arxiv.org/abs/2406.04225v1.

[2] B. Helffer, T. Hoffmann-Ostenhof and S. Terracini, *On spectral minimal partitions: the case of the sphere*, arXiv:0903.3326v1 (2009). https://arxiv.org/abs/0903.3326v1.

[3] N. Soave and S. Terracini, *Liouville theorems and 1-dimensional symmetry for solutions of an elliptic system modelling phase separation*, Advances in Mathematics **279** (2015), 29–66. doi:10.1016/j.aim.2015.03.015. Source version: https://arxiv.org/abs/1404.7288v2.

[4] NIST, *Digital Library of Mathematical Functions*, equations 10.47.3 and 10.49.3. https://dlmf.nist.gov/10.47.E3; https://dlmf.nist.gov/10.49.E3.

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